

Africa Growth Explorer

Predicting next-year GDP per capita growth across African economies from World Bank development indicators

THE QUESTION & THE ANSWER - STATED FIRST

Can recent development indicators predict near-term growth?

No. The model does no better than guessing the historical average. The contribution is a build process careful enough to establish that rather than hide it.

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"I'll give you the conclusion first, because it shapes everything after it. The model does no better than guessing the average and I can show you why that's the correct answer rather than a failed one."

The task & why it matters

THE TASK

Use this year's development indicators to predict next year's GDP per capita growth, across the **54 African UN member states**.

WHY IT'S WORTH DOING CAREFULLY

Development analysts need screening tools. But a growth prediction is only useful if you can state how much better it is than "predict the average" – and that comparison is the step most portfolio projects skip.

FRAMING

- **Not:** a tool that tells you what causes growth
- **Is:** a tool that finds patterns, plus a build process that refuses to ship a model which fails to beat the obvious alternatives
- **Users:** analysts, researchers, policy teams, NGOs

The bar, set before any modelling: Beat all three trivial rules – predict the average, predict this year's growth, predict each country's own average – on the tuning data. Then check on untouched data whether the winning margin is big enough to be real. A model that cannot clear that bar does not ship.

World Bank WDI, 2000–2024

52

Countries in data
(54 configured; 2 missing this vintage)

1,205

Usable country–year rows
(1,300 total, paired with next–year growth)

14

Indicators across 6 themes
(13 report often enough to keep)

SOURCE

World Bank **World Development Indicators (WDI)**, 2000–2024. Public, no licence restriction, cited in [data/README.md](#) . Mauritius and Sudan are missing from this vintage and arrive with the next refresh – stated here rather than rounded up to "54".

WHAT THE MODEL PREDICTS

Indicators at year t → Growth at year t+1
Ghana 2019 indicators → Ghana 2020 GDP per capita growth

HOW THE YEARS WERE DIVIDED – NOTE THE TARGET YEARS RUN ONE LATER

PORTION	INPUT YEARS	GROWTH YEARS PREDICTED	N
Training	2000–2017	2001–2018	905
Tuning	2018–2020	2019–2021 (includes the COVID crash)	150
Testing	2021–2023	2022–2024 (untouched, opened once)	150

"The years being predicted matter more than the input years. Tuning covers the COVID crash; testing covers the recovery. That mismatch forces a design decision on the next slide."

The build – `scripts/finalize_model.py`

Raw WDI download

- select African countries by explicit country code
- build the country-year table, resolve duplicates
- attach each row's following-year growth
- drop indicators reporting under 60% of the time,
measured on TRAINING rows only
- tune settings using training years only, by fitting on an early stretch and checking the years just after, then widening:
trees depth {2,3} × learning rate {0.01,0.03,0.05} × rounds {100,200}
Ridge penalty strength {1 ... 3000}
- score the best of each family on the TUNING data
- CHECK AGAINST THE TRIVIAL RULES - fail ⇒ nothing is saved, build exits
- retrain on training data only, as decided in advance
- OPEN THE TEST SET, exactly once: score, and test whether the margin is real
- stamp every saved file with a data fingerprint, git commit, library versions

WHAT THE DATA SHOWED

- Development levels – electricity, internet, urbanisation, life expectancy, GDP per capita – move together, correlating at **0.45–0.71**. Five measurements of roughly the same underlying thing.
- Current-year growth barely moves with any of them (correlations of **0.13 or less**)
- Missing data clusters in particular countries – so the coverage test runs on training rows only

MODELS COMPARED

- **Ridge regression** – straight-line model whose coefficients carry direction
- **Gradient-boosted trees** – deployed; won on tuning data and beat the trivial rules

THE THREE TRIVIAL RULES

Predict the average · predict this year's growth · predict each country's own past average (using only years already seen)

"I decided how to retrain before opening the test set. The tuning years contain the COVID crash and the test years are the recovery – train across that break and predictions come out systematically low. I locked in training-only beforehand, and I report the alternative so you can see the cost."

Test set – growth years 2022–2024, n=150

Lower MAE and RMSE are better; both are average error sizes in percentage points.

MODEL	MAE	RMSE	R ²	DIRECTION RIGHT	ALWAYS-“UP” SCORES	GAIN
Predict the average	1.90	2.84	-0.00	80.7%	80.7%	0.0 pts
Predict this year's growth	2.23	4.52	-1.54	77.3%	80.7%	-3.3 pts
Predict country's own average	1.94	2.88	-0.03	78.0%	80.7%	-2.7 pts
Gradient-boosted trees (deployed)	1.82	2.79	0.03	80.7%	80.7%	0.0 pts

The model beats the predict-the-average rule by 0.07 percentage points.

Resample the 150-row test set 5,000 times and that margin runs from 0.04 worse to 0.19 better.

A range that includes losing means the lead is not real. The two perform the same. That is the finding.

"The model wins by seven hundredths of a point. But there are only 150 rows in the test set, so I resampled it five thousand times to see how stable that is. On a good number of those resamples the model actually loses. When your margin includes losing, you don't have a margin – you have noise."

TUNING DATA (WHERE THE MODEL WAS CHOSEN)

Ridge 4.00 · trees **3.89** · predict-the-average 4.04 – the trees cleared the bar by 3.7%, a margin well inside noise at 150 rows. The same non-result shows up here.

READING THE DIRECTION FIGURE HONESTLY

The model calls direction right **80.7%** of the time. But growth was positive in 80.7% of test years – so always saying "up" scores exactly the same. Scoring up-years and down-years separately gives **51.3%** – a coin flip. R² is +0.03: a model that has correctly learned to predict the mean and stop.

"If I put 80.7% on a slide with no context, it looks like the strongest number in the deck. It's the weakest. It's just how often growth happened to be positive."

Five things the data established

01 / 05

Only 2 of 14 indicators made a reliable difference

Shuffle one indicator's column at random, re-score, and see how much worse the model gets:

- GDP per capita **0.046** [0.018, 0.085]
- Population growth **0.017** [0.000, 0.037]
- For the other twelve – including electricity and inflation – the range of plausible values includes zero, meaning the model was not really using them

Scale check: scrambling the strongest indicator worsens predictions by 0.046, against typical errors near 3.9 – roughly one percent. Real enough to measure, far too small to act on.

02 / 05

This measures size, never direction

There is no "positive driver / negative driver" column, because shuffling a column tells you how much the model leaned on an indicator, not which way it pushed. An earlier draft had that column; it was a category error and was removed.

03 / 05

Even the straight-line structure is faint

Ridge settled on the strongest penalty in the grid – 3000, the largest value offered. Pulling predictions as hard as possible toward the average is the best a straight-line model can do here. Every coefficient is 0.14 or smaller.

04 / 05

The big misses are events, not model failure

Every worst case is a shock year: Libya 2021 (13.15 pp off), Cabo Verde 2021 (12.53 pp), Equatorial Guinea 2022 (11.01 pp). Conflict, tourism collapse, oil. No set of annual indicators anticipates these.

05 / 05

Why believe this result

The same dataset previously produced a "working" model that was in fact worse than a constant – its test error was nearly double that of simply predicting the average – because the winner had been picked using the test set, and the report's statistics were invented. Fixing the process removed the mirage. This is what the honest machinery reports.

Four pages: Aim = decision support

Page 1 Overview

The problem, the data, the model, the headline result with its caveat stated up front, and the reminder that none of this shows cause and effect.

Page 2 Explore Africa

Country selector, growth over time (what happened vs what the model was asked to predict), indicator trends, regional comparison.

Page 3 Model Performance

The "this margin is not real" banner; how the model scores against the trivial rules on both test and tuning data, always showing what always-saying-"up" would score; where predictions land against reality; the errors; which indicators the model actually used, with the ones it ignored listed separately; the Ridge direction panel; results year by year.

Page 4 Scenario Explorer ← the decision-support feature

- Pick a country and year → its actual indicator values load, with a notice if any were filled in
- Sliders stay inside the range the model trained on, and warn past the edges – inflation warns above ~49.5 rather than 92
- If a country's real value sits outside that range, the slider is pulled to the edge **and says so**
- The change table re-runs the model once per indicator, and says plainly that the effects do not add up
- The cause-and-effect warning stays on screen

Note: Narrate from live output only. No figures are scripted here – an earlier draft's illustrative numbers were removed.

Seven recommendations

FOR ANYONE USING THIS TOOL

- 1 **Do not allocate funding based on these predictions.** They cannot be told apart from simply guessing the average. This is the first recommendation because it is the one with consequences.
 - 2 **Use it to compare and describe** development profiles – the Explore page. That is what the data supports.
 - 3 **Treat every scenario change as the model's response,** never as the effect of a policy.
-

FOR THE NEXT ROUND OF WORK

- 4 **Find faster-moving data** – satellite nightlights, port and air-traffic volumes, mobile-money flows, survey expectations – rather than more annual indicators. Tying with the average argues for better signal, not more models.
- 5 **Keep the automatic check against trivial rules, and keep deciding the rules before looking at the data.** That is the reusable asset from this project.
- 6 **Re-download the WDI data** to restore Mauritius and Sudan before quoting country coverage externally.
- 7 **Watch for periods that behave differently.** This result is partly a statement about a 25-year window containing three very different economic eras.

What makes the result trustworthy

01

Nothing from the future leaks backwards: the coverage test runs on training rows only; filling gaps and rescaling happen inside the model, refitted per fold; each row's target comes from the following year within the same country; the test set stays closed until one final scoring pass

02

A model worse than guessing cannot ship – the check runs before anything is saved, and a failure exits with an error and writes nothing

03

Settings genuinely tuned on training years only, fitting on an early stretch and checking the years just after, with the fold-by-fold results committed (`models/cv_results_*.csv`)

04

The retraining policy was fixed in advance, and the alternative is reported as a cost: error 2.03, predictions 0.86 points low

05

The headline comparison was tested for significance, and reported whatever it said

06

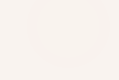
Every saved file records its origin: data fingerprint, git commit, library versions, which years went where. Two runs produce identical numbers

07

86 passing tests, including deliberate sabotage: corrupt the test answers and model selection stays byte-for-byte identical; force the trivial-rule check to fail and confirm nothing gets saved

08

Notebooks run start to finish, with no hardcoded paths, loading the shipped model rather than picking a new one



Three principles to take away

1

A result you can defend beats a win you can't. The process is the contribution.

2

Make your build check the obvious alternatives – "does this beat what it replaces?" belongs in the code, not the discussion section.

3

Generate every number in every document from the saved model, so the write-up cannot drift from what was actually built.



GitHub – code, saved model, tests, executed notebooks, full report



App – verified locally via `streamlit run app.py` ; no public URL (README → Deployment)



Report – `reports/capstone_report.md`



Contact – theresia.saumu@gmail.com

Anticipated questions

Why trees over Ridge?

On the tuning data the best tree model missed by 3.89 points against Ridge's 4.00, and it beat the trivial rules.

Is tying with the trivial rule a failure?

No. It is a measured, tested statement about what this data can and cannot predict, produced by a process with no way to fake a result.

Why did early stopping matter?

The previous version overfit because scikit-learn's automatic setting switches itself off below 10,000 samples, and there are 905 training rows. Setting it explicitly bounds the model – it now stops at 45 rounds out of 200.

Why fix the retraining policy in advance?

The tuning years contain the COVID crash. Training across it pushes predictions systematically low: error 2.03 and 0.86 points low, against 1.82 and 0.08 high for training-only.

Why not tune harder?

Hunting for settings that score well on the test set is exactly how the previous version shipped a bad model. The grid is small, fixed in advance, and scored only within training years.

How often retrain?

Annually, as WDI updates. The trivial-rule check and the significance test must pass each time.

Can this inform policy?

As background screening only. Cause-and-effect claims need experiments or natural experiments.

What would change your mind?

Faster-moving data, or a test window that isn't a pandemic recovery. Both are in Recommendations.

● APPENDIX - ONLY IF ASKED

Supporting material

A1 Fold-by-fold tuning results (`models/cv_results_*.csv`)

A2 How the ranges around each number were computed, and the head-to-head significance test (report §7)

A3 The largest misses, every row checked against the frozen predictions (report §7)

A4 Which indicators were kept and why – secondary school enrolment was dropped for reporting only 59.9% of the time